

HANFORD, WA, USA

WMO: 727840

Lat: 46.567N

Lon: 119.600W

Elev: 732

StdP: 14.31

Time Zone: -8.00 (NAP)

Period: 86-95

WBAN: 94187

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.7	15.2	-0.5	5.5	13.6	7.2	8.2	20.3	29.9	49.6	24.6	44.3	6.1	20	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.2	100.7	67.3	97.4	65.5	93.8	64.2	68.3	97.6	66.5	93.9	65.2	91.3	7.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	73.7	72.2	55.6	67.6	73.6	53.6	62.9	74.0	32.7	97.3	31.3	94.2	30.3	91.0	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.2	20.9	18.5	DB	4.6	105.5	8.5	3.4	-1.5	107.9	-6.5	109.9	-11.2	111.9	-17.4	114.3	(4)
(5)			WB	4.2	69.4	8.0	4.5	-1.6	72.7	-6.2	75.3	-10.7	77.9	-16.5	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	33.8	38.4	46.9	55.3	63.5	71.1	76.1	76.5	67.9	55.2	42.2	32.6	(6)	
	DBStd	17.29	8.79	9.66	6.29	6.34	7.94	7.80	7.19	6.59	6.95	7.52	7.98	9.05	(7)	
	HDD50	1810	502	332	130	19	1	0	0	0	0	30	255	541	(8)	
	HDD65	4784	966	744	560	298	128	26	3	3	51	315	685	1005	(9)	
	CDD50	3653	1	8	35	178	419	632	810	821	538	191	19	1	(10)	
	CDD65	1154	0	0	0	7	82	208	348	359	139	11	0	0	(11)	
	CDH74	15244	0	0	4	140	1145	2713	4611	4632	1830	169	0	0	(12)	
	CDH80	7759	0	0	0	27	513	1356	2541	2491	800	31	0	0	(13)	
	Wind	WSAvg	7.8	6.7	7.2	7.8	8.8	9.2	9.3	8.3	7.4	6.7	7.3	6.0	(14)	
	Precipitation	PrecAvg	7.3	0.9	0.6	0.5	0.6	0.6	0.3	0.2	0.2	0.7	0.9	1.1	(15)	
(16)		PrecMax	12.5	2.4	1.1	1.1	1.7	1.4	2.0	1.7	1.0	0.8	1.3	3.6	(16)	
(17)		PrecMin	4.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.2	(17)	
(18)		PrecStd	1.9	0.6	0.3	0.4	0.4	0.5	0.4	0.2	0.2	0.4	0.5	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	65.0	73.1	83.7	97.9	103.3	106.0	103.4	96.7	83.8	65.3	56.9	(19)	
		MCWB	49.6	53.7	53.3	60.5	66.6	68.8	69.7	68.4	64.8	60.2	54.0	47.7	(20)	
	2%	DB	54.3	60.4	67.5	77.9	91.6	96.7	101.0	100.0	92.7	78.9	61.3	52.5	(21)	
		MCWB	45.9	49.8	51.3	57.3	63.2	65.6	66.8	67.1	63.0	58.7	51.3	44.9	(22)	
	5%	DB	50.0	56.0	62.9	73.6	85.9	92.4	97.2	96.8	88.6	74.3	57.8	48.7	(23)	
		MCWB	43.0	46.2	49.3	54.3	60.7	63.8	65.1	65.6	61.8	56.4	49.2	42.2	(24)	
	10%	DB	45.6	51.8	59.5	69.4	80.1	88.0	93.3	92.5	84.6	69.5	54.0	44.5	(25)	
		MCWB	40.2	43.6	47.2	52.5	58.1	61.9	64.1	63.9	60.2	53.6	46.0	39.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	55.2	55.0	61.8	69.4	69.3	70.0	70.1	65.8	62.0	55.0	49.0	(27)	
		MCDB	58.1	63.3	68.9	80.7	94.7	101.9	103.1	101.2	94.4	81.4	62.7	55.2	(28)	
	2%	WB	46.4	50.1	52.8	58.9	63.8	66.5	67.9	68.1	64.0	59.4	52.6	45.8	(29)	
		MCDB	53.3	59.3	63.6	74.0	87.8	94.7	97.4	96.7	89.9	77.0	59.9	51.5	(30)	
	5%	WB	43.7	46.7	50.7	56.1	61.3	64.5	66.3	66.5	62.4	57.2	49.9	42.9	(31)	
		MCDB	49.5	54.3	61.3	70.8	82.9	89.8	93.9	93.2	86.7	71.4	56.9	47.7	(32)	
	10%	WB	40.8	44.4	48.4	53.5	59.1	62.6	64.8	65.0	60.9	54.9	46.9	40.1	(33)	
		MCDB	44.6	50.6	58.0	67.1	78.5	85.5	90.6	90.2	83.3	67.2	53.0	44.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	17.1	21.9	24.0	25.5	26.1	27.3	28.2	28.8	24.2	15.6	11.1	(35)	
		MCDBR	18.5	23.5	27.2	30.0	32.7	31.0	31.4	31.6	34.5	29.0	17.9	17.6	(36)	
		MCWBR	11.5	14.2	15.1	14.6	13.0	12.5	11.5	11.8	14.1	13.7	10.7	11.8	(37)	
	5% WB	MCDBR	18.1	20.6	24.8	27.2	30.7	29.3	30.3	29.5	32.1	27.8	16.7	16.5	(38)	
		MCWBR	11.5	13.0	14.4	13.4	12.8	12.2	11.5	11.5	13.2	13.2	10.7	11.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.286	0.290	0.299	0.331	0.346	0.346	0.347	0.344	0.331	0.309	0.293	0.282	(40)	
	taud		2.548	2.566	2.563	2.439	2.415	2.440	2.442	2.443	2.476	2.538	2.544	2.539	(41)	
	Ebn at Noon		263	285	298	294	292	291	289	286	281	273	257	251	(42)	
	Edh at Noon		20	23	27	34	36	35	35	33	29	23	19	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		362	714	1130	1676	2034	2270	2428	2035	1505	928	478	300	(44)	
	RadStd		47	93	76	96	119	111	84	110	83	73	45	36	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+76	

Nomenclature: See separate page